BAG ML Team

Problem 2

a) $K^T Q = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = S$

S is 2x2. The i,j entry represents the dot product of the key vector at token i and the query vector of token j. It is the raw unnormalized attention score

b) $\hat{S} = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$

c) $\begin{pmatrix} 0.5 & 0.3302 \\ 0.5 & 0.6698 \end{pmatrix} = \text{softmax}(\hat{S}) = A$

d) $Z = VA = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} .5 & .3302 \\ .5 & .6698 \end{pmatrix} = \begin{pmatrix} 1 & .6604 \\ 1.5 & 2.0094 \end{pmatrix}$

e) The first column of A is $\begin{pmatrix} .5 \\ .5 \end{pmatrix}$ which represents the attention of query one. It distributes attention evenly, meaning that it attends/pulls from both token 1 and 2 equally

Problem 3

a) numerators: e^2, e^1, e^0

sum: $e^2 + e + 1$

$$\text{softmax}(s) = \begin{pmatrix} \frac{e^2}{e^2+e+1} \\ \frac{e}{e^2+e+1} \\ \frac{1}{e^2+e+1} \end{pmatrix} \Rightarrow \frac{e^2+e+1}{e^2+e+1} = 1 \checkmark$$

b)

$$\text{softmax}(s') = \begin{pmatrix} \frac{e^4}{e^4+e^2+1} \\ \frac{e^2}{e^4+e^2+1} \\ \frac{1}{e^4+e^2+1} \end{pmatrix}$$

The distribution becomes more extreme, with 4 taking significantly more of the total probability mass

Problem 3c)

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Problem 4)

a)

b)

c) still

d) Z=

Z=

Problem 3c)

If the raw attention scores get too high, softmax output basically turns into a one-hot vector. This makes the gradients vanish and become extremely small, stalling training. This is why we divide by $\sqrt{d_k}$

Problem 4)

a) $d_h(X) \to 512 \times 6$

for $Q = \underbrace{(W^{(Q)})^T}_{64 \cdot 6} \underbrace{X}_{512 \cdot 6}$

$512 \cdot 64$

$64 \cdot 6$

b) $K = 64 \cdot 6$

$Q = 64 \cdot 6$

$K^T Q = 6 \cdot 6$

$6 \cdot 64 \quad 64 \cdot 6$

c) still 6×6 (we just scale by $\sqrt{d_k}$)

d) $Z = VA$

$64 \cdot 6 \quad 6 \cdot 6$

$Z = 64 \cdot 6$

Because every individual value vector has a height of 64, their sum is still 64 units tall.

Problem 5

a) $\underbrace{256 \times 64}_{\text{parameters per matrix}} \times \underbrace{3}_{\text{3 matrices}} \times \underbrace{4}_{\text{4 heads}} = 196{,}608$

b) $\dim(W^O) = (h \cdot d_v) \times d_{model}$

$= 4 \cdot 64 \cdot 256 = 65{,}536$

$196{,}608 + 65{,}536 = 262{,}144$ per layer

c) $6 \cdot 262{,}144 = 1{,}572{,}864$

A total per layer:

$$(3 \cdot \underbrace{h \cdot d_{model} \cdot d_k}_{d_{model}^2}) + (\underbrace{h \cdot d_v}_{d_{model}} \cdot d_{model})$$

$3 \cdot d_{model}^2 + d_{model}^2 = 4(d_{model})^2$

parameter count scales quadratically, so doubling d_{model} scales attention parameter count by 4